

Consumer Behavior Analysis

1. Executive Summary

This project delivers a data-driven analysis of consumer shopping transactions to uncover insights into purchasing behavior, product performance, and customer segmentation. By combining Python for data engineering, PostgreSQL for deep querying, and Power BI for interactive visualization, this analysis transforms raw transactional data into actionable business intelligence.

The primary objective is to optimize marketing spend, refine inventory strategies, and enhance customer retention through data-backed decisions.

2. Technical Architecture & Pipeline

The project implements a classic three-tier data analytics pipeline:

[Raw CSV Data] → [Python (Pandas/SQLAlchemy)] → [PostgreSQL DB] → [Power BI Dashboard]

- **Data Extraction & Cleaning (script.ipynb):** Handles initial data ingestion, resolves missing data points (imputing missing review ratings via category medians), normalizes schema features, handles feature engineering (segmenting age into quartiles), and exports the polished dataset.
- **Database Layer (sql_queries.sql):** Provides deep-dive relational analytics across 10 core business questions, mapping complex customer segmentation (New vs. Loyal) and revenue contributions.
- **Visualization Layer (Consumer_Behavior.pbix):** An executive-ready reporting dashboard designed for dynamic filtering and immediate cross-category analysis.

3. Dashboard Insights & Visual Analysis

Based on the executive **Consumer Behavior Dashboard**, the business currently operates with the following baseline metrics:

High-Level KPIs

- **Total Customer Base:** 3.9K unique customers.
- **Average Order Value (AOV):** \$59.76 per transaction.
- **Customer Satisfaction:** A healthy **3.75 / 5.00** average review rating.

Key Visual Breakdown

Visual Metric	Observed Trend / Finding	Business Implication
Subscription Status	73% Non-Subscribers vs. 27% Subscribers.	A significant portion of the user base is transactional; driving subscription conversions is a massive growth lever.
Category Performance	Clothing is the clear market leader in both total Revenue and Sales Volume, followed by Accessories, Footwear, and Outerwear.	Inventory allocation and prime digital shelf space should heavily favor Clothing and Accessories.
Purchases by Age Group	Young and Middle-Aged segments lead total transaction volume, though engagement remains relatively stable across all four cohorts (including Adults and Seniors).	Marketing campaigns should remain broad but prioritize youthful lifestyle messaging to maximize ROI.

UI Observation: The bottom two bar charts are both titled "*Purchases by Age Group*". The left chart tracks total purchase volumes (scaled up to 62K metrics), while the right chart isolates explicit transaction/customer counts (ranging from 942 to 1028).

4. Strategic Database Deep Dives

The SQL analysis script bridges the gap between high-level visual charts and granular data patterns. Key query focuses include:

- Gender Revenue Disparity:** Measuring total capital spend differences between demographic segments.
- High-Value Discount Users:** Identifying cost-conscious power buyers who utilize promotions but maintain an above-average spend.
- Shipping Economics:** Comparing performance and average spend between standard and express shipping tiers to evaluate logistical margins.
- Retention Health:** Classifying users into *New*, *Returning*, and *Loyal* tiers based on historical touchpoints.

5. Strategic Recommendations & Optimization Plan

Data & Technical Fixes

- **Fix Dashboard Dual Titles:** Update the title of the bottom-right chart to "**Customer Count by Age Group**" or "**Transaction Count by Age Group**" to eliminate ambiguity.
- **Correct SQL Revenue Aggregations:** As noted in the data audit, ensure Query #10 strictly utilizes SUM(purchase_amount) rather than COUNT to prevent misleading revenue share conclusions.
- **Database Schema Alignment:** Ensure the database table name is cleanly migrated to customer to resolve the current constomer typo in the pipeline script.

Business & Marketing Actions

- **Monetize Non-Subscribers:** With 73% of customers currently sitting outside the subscription framework, launch an incentive campaign (e.g., *"Free shipping on your next order if you subscribe today"*) targeting the high-volume Clothing and Accessories shoppers.
- **Targeted Loyalty Perks:** Use the SQL-derived customer tiers (New vs. Loyal) to build automated email triggers via Outlook. Offer exclusive early-access perks to the "Loyal" group to maximize lifetime value (LTV).